

Phase offset ambiguities in multi-echo field mapping

What can be recovered from wrapped phase, and what cannot

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1 Why the phase offset matters

Multi-echo field mapping reads the field off the **rate** at which phase accumulates across echoes. That rate is the quantity of interest; the constant it accumulates from is a nuisance. Every voxel carries a receive-coil phase offset θ — present already at $t = 0$, identical on every echo, and set by coil geometry rather than by the field.

θ cannot simply be ignored. It varies rapidly in space, steeply near coil elements and discontinuously between channels, so the measured phase is not spatially smooth even where the field is. Spatial unwrapping assumes neighbouring voxels differ by less than half a turn, and on raw phase that assumption fails. Removing θ first restores the smoothness that everything downstream depends on.

The difficulty is that θ is never measured. It has to be inferred from the same wrapped phase it corrupts, and that inference is not unique: several distinct (θ, f) pairs explain the same data. Some of the resulting ambiguity can be resolved from additional echoes, and some of it provably cannot.

This note draws that line. It enumerates the ways an estimate can be wrong, shows which of them leave a trace in the data, and derives what to do about each. Everything below follows from the signal model.

2 The measurement

2.1 Signal model

Acquire E echoes at times $t_0 < t_1 < \dots < t_{E-1}$. Field evolution is linear in time, so before wrapping, the phase at echo e would be $\theta + 2\pi f t_e$, with f the **field offset** in Hz — the quantity wanted — and θ the coil offset described above.

What is recorded is that value wrapped into a single turn. Writing $W(x) = \arg(e^{ix})$ for wrapping to $(-\pi, \pi]$,

$$\varphi_e = W(\theta + 2\pi f t_e). \quad (1)$$

The entire difficulty comes from W . Without it, two echoes would determine θ and f exactly by fitting a straight line. With it, the line is observed only modulo 2π , and infinitely many (θ, f) pairs remain compatible with the data.

2.2 Evenly spaced echoes

An echo-planar train produces echoes at constant spacing,

$$t_e = \text{TE}_0 + e \cdot \Delta \text{TE}, \quad e = 0, 1, \dots, E-1, \quad (2)$$

with TE_0 the first echo time and ΔTE the spacing. The single most important derived quantity below is their ratio,

$$k = \frac{\text{TE}_0}{\Delta \text{TE}}, \quad (3)$$

dimensionless and fixed by the protocol. Two field scales also recur:

- **Wrap spacing** $1/\Delta \text{TE}$ (Hz) — how far apart the candidate field solutions sit.
- **Half-wrap** $1/(2\Delta \text{TE})$ (Hz) — the midpoint between neighbouring candidates, and therefore the decision boundary for any rule preferring the smallest field.

2.3 Removing the offset

Because θ is common to every echo, it cancels in a difference. The phase difference of the first two echoes removes the coil contribution and leaves a quantity proportional to the field:

$$\Delta\varphi = W(\varphi_1 - \varphi_0) \approx W(2\pi f \Delta \text{TE}). \quad (4)$$

This is still wrapped, but a difference of **neighbouring** echoes varies slowly in space, so a region-growing algorithm can restore the missing turns and produce an unwrapped difference $\Delta\varphi_{\text{uw}}$. Extrapolating back to $t = 0$ then recovers the offset:

$$\hat{\theta} = W(\varphi_0 - k \cdot \Delta\varphi_{\text{uw}}). \quad (5)$$

Spatial unwrapping recovers **relative** phase between voxels. It cannot know the absolute number of turns, so it determines $\Delta\varphi_{\text{uw}}$ only up to a global additive constant $2\pi M$ for some unknown integer M . Pinning down M is the decision this note is about.

3 A taxonomy of wrong answers

Suppose some estimate $(\hat{\theta}, \hat{f})$ has been produced. To classify how it can be wrong, remove the estimated offset from the data and inspect what remains:

$$\psi_e = \varphi_e - \hat{\theta}. \quad (6)$$

Were the estimate perfect, ψ_e would be exactly proportional to t_e : a straight line through the origin whose slope is the field. So fit

$$\psi_e \approx c + s \cdot t_e \quad (7)$$

and read off two numbers — the **intercept** c and the **slope** s . Every possible error is a displacement in this (c, s) plane, and exactly three cases are of interest.

- A. **Both move together.** Offset and field are both wrong, in a matched combination that leaves ψ_e unchanged. Invisible.
- B. **The offset moves alone.** c shifts while s stays correct. Visible as a line that misses the origin.
- C. **The field moves alone.** s shifts while c stays exactly zero. Visible as a wrong slope — but **not** as a wrong intercept.

Each is taken in turn below. The taxonomy matters because a test that measures c finds case B and is blind to A and C **by construction** — not through any weakness of implementation.

3.1 Ambiguity A: an exact symmetry of the data

Proposition 1. For evenly spaced echoes and any integer N , define

$$\theta' = \theta - W(2\pi kN), \quad f' = f + \frac{N}{\Delta \text{ TE}}. \quad (8)$$

Then (θ', f') reproduces Equation 1 **identically**: $\varphi'_e = \varphi_e$ for every echo.

Proof. Using $W(2\pi kN) \equiv 2\pi kN \pmod{2\pi}$,

$$\begin{aligned} \theta' + 2\pi f' t_e &\equiv \theta - 2\pi kN + 2\pi(f + N/\Delta \text{ TE})t_e \\ &= \theta + 2\pi f t_e + 2\pi N(t_e/\Delta \text{ TE} - k). \end{aligned} \quad (9)$$

For evenly spaced echoes $t_e/\Delta \text{ TE} = k + e$, so the bracket equals the integer e and the term vanishes under W . ■

Offset error and field error are locked together here: shift the field by one wrap spacing and the offset compensates exactly. The recorded data is **bit-identical** — not approximately, but exactly.

No statistic computed from the phase can separate these alternatives, because there is nothing to separate: they are the same measurement. Resolving A requires information from outside the data, which is to say a prior.

3.1.1 Why more echoes do not help

It is natural to expect a five-echo acquisition to constrain the field better than a two-echo one. For this question it does not. Prop. 1 holds for all echoes simultaneously, and its period $1/\Delta$ TE depends only on the spacing, which every pair shares. All pairs therefore alias at the same period and agree with one another.

Unequal spacing would break the tie — two pairs whose aliases have incommensurate periods are simultaneously satisfied only by the true field — but an EPI echo train produces equal spacing naturally, so that escape is not usually available.

3.2 Ambiguity B: the offset moves alone

Now suppose spatial unwrapping settled on the wrong number of turns, so $\Delta\varphi_{\text{uw}}$ is off by $2\pi M$, while the field has already been fixed by an earlier stage and does not follow.

Proposition 2. If $\Delta\varphi_{\text{uw}}$ carries a branch error of M wraps, Equation 5 returns $\hat{\theta}_M = W(\theta - 2\pi kM)$, and the offset-removed phases satisfy

$$\psi_e \equiv 2\pi f t_e + c \bmod 2\pi, \quad c = W(2\pi kM), \quad (10)$$

with c **identical on every echo**.

Proof. Substituting $\Delta\varphi_{\text{uw}} \rightarrow 2\pi f \Delta \text{ TE} + 2\pi M$ into Equation 5 gives $\hat{\theta}_M = W(\theta - 2\pi kM)$. Then $\psi_e \equiv (\theta + 2\pi f t_e) - (\theta - 2\pi kM) = 2\pi f t_e + 2\pi kM$, and $2\pi kM \equiv c \bmod 2\pi$. ■

Comparing against Equation 7: the slope is untouched and the intercept has moved to $c = W(2\pi kM)$. Because c is the **same on every echo**, it does not resemble noise and does not average away — it is a rigid vertical displacement of the whole line.

This is the one detectable case, and the reason is worth stating plainly: offset and field have become mutually inconsistent, and consistency between them is something the data can check.

3.3 Ambiguity C: the field moves alone

The third case is easy to overlook, because it does not originate in the offset estimate at all.

Multi-echo unwrappers commonly unwrap a single **template** echo spatially and derive the remaining echoes by temporal unwrapping relative to it, each echo referenced to the template scaled by its own echo time. A whole-turn error in the template then propagates to echo e scaled by t_e/t_0 — that is, **proportionally to echo time**.

A displacement proportional to t_e is, by Equation 7, a pure change of slope:

$$\Delta s = \frac{2\pi}{t_0}, \quad \Delta c = 0. \quad (11)$$

The intercept is exactly zero here. A test built on the intercept cannot see ambiguity C at all — not poorly, but not at all. It surfaces instead as a field error of $1/t_0$ Hz, a different size from the $1/\Delta$ TE of ambiguity A, which is the practical way to tell the two apart.

4 Detecting ambiguity B

By Prop. 2 a branch error of M wraps places the constant $c = W(2\pi kM)$ on every echo. Fitting the line of Equation 7 and reading its intercept therefore measures the error directly. Because the effect is a single global constant rather than a per-voxel one, the intercept can be aggregated over many voxels, suppressing noise without diluting the signal.

The procedure follows: for each candidate M , reconstruct the offset it implies, remove it, fit, and record $|c|$. The correct branch extrapolates through the origin; a wrong one does not.

4.1 The rejection scale is analytic

Any test needs a scale — how large must $|c|$ be before a candidate is rejected? Here the model supplies it rather than the user. Candidate intercepts lie on a lattice of spacing

$$\Delta c = |W(2\pi k)|, \quad (12)$$

a function of the echo times alone. Classifying against $\Delta c/2$ is nearest-neighbour assignment on that lattice — the midpoint between two adjacent possibilities — rather than a tuned threshold.

4.2 Where the test loses power

Integer k . If k is a whole number then $c = W(2\pi kM) = 0$ for every M , so the intercept carries no information. But by Prop. 2 the recovered offset $\hat{\theta}_M = W(\theta)$ is **also unchanged**, so no choice of branch can affect the output. The degeneracy is harmless, and Equation 12 returning zero is the correct signal to do nothing.

Away from integer k the achievable separation varies smoothly, peaking near $k = 2/3$ and shrinking to either side. For small k in particular, several candidates can be exactly through-origin at once: the alias of Prop. 1 becomes reachable **within the candidate set**, and the intercept cannot rank the survivors. Since an EPI train has TE_0 shorter than the echo spacing, $k < 1$ always, so this regime is not exotic.

5 Choosing among indistinguishable candidates

Once more than one candidate is through-origin the data has been exhausted and something else must decide. Two facts constrain what that can be.

A consistency test alone is insufficient. Tied candidates have equal intercepts to within numerical noise. Ranking them is ranking noise.

A field-magnitude prior alone is insufficient. A rule preferring the smallest $|f|$ is **symmetric about zero**: candidates at $+\delta$ and $-\delta$ are equidistant and cannot be separated. Such pairs do arise, because re-wrapping inside the unwrapper can map candidates M and $-M$ to equal $|f|$.

The two are complementary in a precise sense. The intercept is blind to a **sign-preserving** change of the field — that is ambiguity C, and the alias direction of A. The magnitude prior is blind to a **sign-reversing** one. Together they cover both, which is why a workable rule needs both stages.

The resulting rule: reject candidates whose intercept is inconsistent; if one survives, take it; if several survive, apply the prior among them; if none survives, change nothing, since a failed fit is not evidence for any particular candidate.

6 The prior and how to estimate it

6.1 What the prior actually asserts

Before imaging, the scanner sets its demodulation frequency to the centre of the water resonance over the shim volume. The physical content of the prior is therefore:

The **bulk-tissue** field is near zero. Not the mean field, and not the median field over whatever mask happens to be convenient.

The distinction is not pedantry. The field distribution over a brain is asymmetric: a narrow peak at the bulk-tissue value, plus long tails from air–tissue interfaces, sinuses, signal dropout and mask edges, where susceptibility gradients are large. Any statistic that is a **quantile of the whole distribution** — the median included — is displaced toward whichever tail is heavier.

6.2 Why the median is the wrong estimator

Model the distribution as a mixture of a narrow bulk component and a broad tail component,

$$p(f) = (1 - \varepsilon) \cdot p_{\text{bulk}}(f) + \varepsilon \cdot p_{\text{tail}}(f). \quad (13)$$

The centre of p_{bulk} is what the shim actually set. For small ε the **mode** of p is a consistent estimator of that centre; the **median** of p is not, and its bias grows with both the tail fraction ε and the tail asymmetry. Two consequences follow, and they are the same defect seen twice.

- **Enlarging the mask makes the estimate worse.** Dilating a brain mask admits precisely the voxels populating p_{tail} , raising ε . A median over a generous mask is more biased than the same median over a conservative one — the opposite of the usual intuition that more data is better.
- **A bias of a fraction of a wrap is decisive.** The prior compares $|\hat{f}|$ against the half-wrap $1/(2\Delta \text{ TE})$. Estimator bias adds directly to the distance from that boundary, so a subject whose true bulk field lies near the half-wrap can be pushed across it by bias alone. The choice then turns on an artefact of the statistic rather than a property of the data.

Weighting voxels by squared magnitude helps, since low-signal voxels are over-represented in the tails, but it is a mitigation rather than a fix: the estimand is still a quantile of the mixture.

6.3 Estimating the mode without a tuning parameter

The mode is the right estimand, but histogram and kernel estimators need a bin width or a bandwidth — reintroducing exactly the kind of tuned constant a principled rule should avoid. The **half-sample mode** needs neither.

Sort the samples. Repeatedly locate the shortest contiguous interval containing half the remaining points, and discard everything outside it. Stop when three or fewer points remain, and return their mean.

The justification is direct: per unit of value, samples are densest where the density is highest, so the shortest half-interval brackets the peak, and recursing zooms in on it. Formally, with order statistics $x_{(1)} \leq \dots \leq x_{(n)}$ and $h = \lceil n/2 \rceil$, the retained window is

$$\left[x_{(i^*)}, x_{(i^*+h-1)} \right], \quad i^* = \arg \min_i (x_{(i+h-1)} - x_{(i)}), \quad (14)$$

and all $n - h + 1$ candidate widths follow from one vectorised difference of two offset slices of the sorted array.

Properties.

- **Parameter-free.** No bin width, no bandwidth, no threshold.
- **Cost.** $O(n \log n)$, dominated by the initial sort; subsequent passes are linear in a geometrically shrinking array. Cheaper than a weighted median, which must sort a weight array as well.
- **Convergence.** Near a smooth unimodal peak, halving the sample halves the window width, so the interval contracts geometrically and the result is insensitive to how many passes are taken.
- **Weight-independent.** Peak location does not depend on how samples are weighted, so none of the weighting choices a median requires are inherited.

Assumptions. Unimodality — with a genuinely bimodal distribution the method converges on the denser mode. Ties in the arg-min are resolved arbitrarily; on continuous data they have measure zero, on quantised or very small samples they do not. And the sample must be large enough to resolve the density at all.

7 What remains unresolvable

Three limits are properties of the acquisition rather than of any algorithm.

Ambiguity A cannot be corrected. By Prop. 1 the data is invariant under the alias, so when the true field genuinely exceeds the half-wrap, no rule recovers it: the prior folds it back, and that is that. Whether the situation is **reachable** is governed by ΔTE , bounded below by the readout duration — echoes cannot be spaced closer than they can be read out — and above by T_2^* , since several echoes must fit before the signal decays. Those bounds confine ΔTE , and so hold the half-wrap threshold above typical shim residuals.

Higher field strength does not simply make it worse. Shim residuals grow with B_0 , which argues for more risk. But T_2^* shortens, forcing ΔTE down and therefore raising the $1/(2\Delta \text{TE})$ threshold. The two effects push the same quantity in opposite directions.

Ambiguity C needs a different detector. A whole-turn error in a template echo produces a pure slope change with exactly zero intercept, so no refinement of an intercept test will find it. Recognising it means looking at the field itself, and its characteristic size is $1/t_0$ Hz rather than the $1/\Delta$ TE of the other cases.

8 Summary

Wrapped phase determines the coil offset and the field only up to a family of alternatives. Sorting them out means asking, of each candidate, whether the offset and the field it implies are mutually consistent.

Three cases exhaust the possibilities. When offset and field are both wrong in a matched way (**A**) the data is unchanged, and only a prior can help. When the offset is wrong alone (**B**) every echo carries the same constant, so extrapolating the echoes back to $t = 0$ exposes it, against a rejection scale $|W(2\pi k)|$ fixed by the echo times. When the field is wrong alone (**C**) the displacement is proportional to echo time — a pure slope change no intercept test can see.

Where several candidates remain consistent, the prior that decides is that **bulk-tissue** field is near zero. That makes the mode of the field distribution, not its median, the correct estimand, and the half-sample mode estimates it without introducing a tuned constant. Consistency and prior are blind to opposite symmetries, so a workable rule needs both.